Instructions on how to set up and create secret keys for Google AI and OpenAI are provided in the appendix.

12.3.1 Architecture overview

Figure 12.2 illustrates the high-level architecture of the system. The integration is organized into five logical layers, each playing a distinct role in the flow between the user, the AI, and external data sources:

- *Browser (Client)*—A React-based chat interface built with the Vercel AI SDK. Users type messages, and the SDK manages streaming responses and conversation state.
- *Next.js API route*—Acts as a broker between the frontend and the AI backend. It forwards user messages to the Vercel AI SDK server utilities, which handle the actual model interaction.
- *Vercel AI SDK*—Connects the Next.js API route with the MCP server. This SDK layer interprets model responses, manages tool calls, and coordinates communication between the LLM and MCP servers.

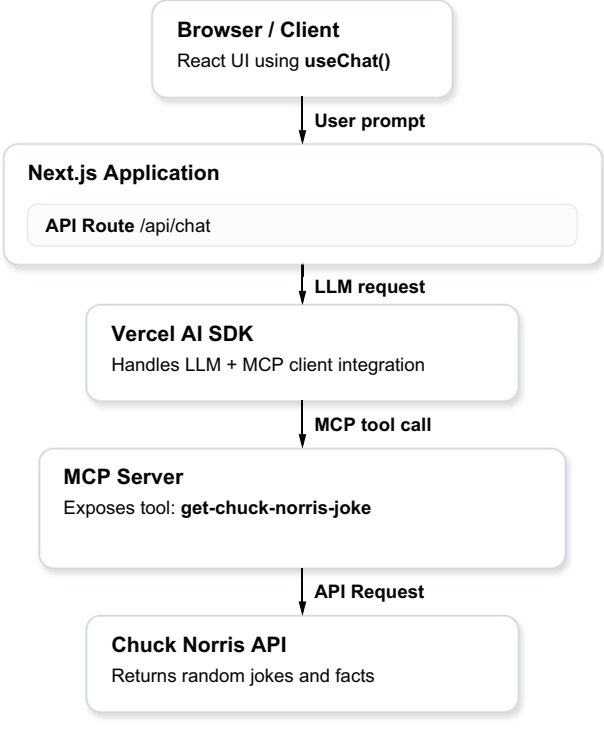


Figure 12.2 High-level architecture showing how a Next.js chat application integrates with the MCP. The Vercel AI SDK communicates with an MCP client, which connects to one or more MCP servers exposing tools. Responses flow back through the client to the API route and are displayed in the chat interface.